

Day 35 - Lagrangian Examples II



$$F = ma$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = 0$$

Announcements

- Homework 8 is "Late" 24 Apr
 - Last Exercise 0: Reflect Learning Outcomes
- Final Project is posted
 - Video Presentations due 27 Apr
 - Computational Essay due 1 May
 - Rubric for both are posted
- No class (20 Apr - 24 Apr) - DC out of country
 - Make appointment for project help (clicker extra credit)

Announcements

Rest of Semester Schedule

- CW16 - Examples of Lagrangian Dynamics (HW8)
- CW17 - Project Prep (DC out of country)
- CW18 - Final Project Due
 - Video Presentations due 27 Apr
 - Computational Essay due 1 May

NO IN-CLASS FINAL EXAM

Clicker Question 35-1a

For a hypothetical system, we have the Lagrangian that depends on two generalized coordinates, $\mathcal{L}(\rho, \phi, \dot{\rho}, \dot{\phi})$.

Here ρ has units of length and ϕ has units of angle.

Which of the following derivatives give the **generalized force** associated with ρ ?

1. $\frac{\partial \mathcal{L}}{\partial \rho}$
2. $\frac{\partial \mathcal{L}}{\partial \dot{\rho}}$
3. $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \rho} \right)$
4. $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\rho}} \right)$
5. None of these

Clicker Question 35-1b

For a hypothetical system, we have the Lagrangian that depends on two generalized coordinates, $\mathcal{L}(\rho, \phi, \dot{\rho}, \dot{\phi})$.

Here ρ has units of length and ϕ has units of angle.

What are the units of the **generalized force** associated with ρ ?

1. Newtons (N)
2. Joules (J)
3. Newton-meters (N m)
4. Meters (m)
5. None of these

Clicker Question 35-1c

For a hypothetical system, we have the Lagrangian that depends on two generalized coordinates, $\mathcal{L}(\rho, \phi, \dot{\rho}, \dot{\phi})$.

Here ρ has units of length and ϕ has units of angle.

Which of the following derivatives give the **generalized momentum** associated with ρ ?

1. $\frac{\partial \mathcal{L}}{\partial \rho}$ 2. $\frac{\partial \mathcal{L}}{\partial \dot{\rho}}$

3. $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \rho} \right)$ 4. $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\rho}} \right)$

5. None of these

Clicker Question 35-1d

For a hypothetical system, we have the Lagrangian that depends on two generalized coordinates, $\mathcal{L}(\rho, \phi, \dot{\rho}, \dot{\phi})$.

Here ρ has units of length and ϕ has units of angle.

What are the units of the **generalized momentum** associated with ρ ?

1. kg m/s
2. kg m²/s
3. kg m²/s²
4. kg/s
5. None of these

Clicker Question 35-1e

For a hypothetical system, we have the Lagrangian that depends on two generalized coordinates, $\mathcal{L}(\rho, \phi, \dot{\rho}, \dot{\phi})$.

Which of the following derivatives give the **generalized force** associated with ϕ ?

1. $\frac{\partial \mathcal{L}}{\partial \phi}$

2. $\frac{\partial \mathcal{L}}{\partial \dot{\phi}}$

3. $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \phi} \right)$

4. $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right)$

5. None of these

Clicker Question 35-1f

For a hypothetical system, we have the Lagrangian that depends on two generalized coordinates, $\mathcal{L}(\rho, \phi, \dot{\rho}, \dot{\phi})$.

Here ρ has units of length and ϕ has units of angle.

What are the units of the **generalized force** associated with ϕ ?

1. Newtons (N)
2. Joules (J)
3. Newton-meters (N m)
4. Radians (rad)
5. None of these

Clicker Question 35-1g

For a hypothetical system, we have the Lagrangian that depends on two generalized coordinates, $\mathcal{L}(\rho, \phi, \dot{\rho}, \dot{\phi})$.

Here ρ has units of length and ϕ has units of angle.

Which of the following derivatives give the **generalized momentum** associated with ϕ ?

1. $\frac{\partial \mathcal{L}}{\partial \phi}$ 2. $\frac{\partial \mathcal{L}}{\partial \dot{\phi}}$

3. $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \phi} \right)$ 4. $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right)$

Clicker Question 35-1h

For a hypothetical system, we have the Lagrangian that depends on two generalized coordinates, $\mathcal{L}(\rho, \phi, \dot{\rho}, \dot{\phi})$.

Here ρ has units of length and ϕ has units of angle.

What are the units of the **generalized momentum** associated with ϕ ?

1. kg m/s
2. kg m²/s
3. kg m²/s²
4. kg/s
5. None of these

Clicker Question 35-2

For the Atwood's machine, M is connected to m by a string of length l . Each mass has a length of string extended as measured from the center of the pulley (R) of y_1 and y_2 , respectively. The string wraps around half the pulley.

Which of the following represents the equation of constraint for the system?

1. $y_1 + y_2 = l - R\phi$

2. $y_1 - y_2 = l + R\phi$

3. $y_1 + y_2 = l - \pi R$

4. $y_1 - y_2 = l + \pi R$

5. None of these

Take the time derivative of the constraint equation. What do you notice? 12 / 15

Clicker Question 35-3

With a Lagrangian of the form $\mathcal{L} = \frac{1}{2}(M + m)\dot{y}_1^2 - (M - m)gy_1$, we can find the **generalized forces** and **generalized momenta**.

$$F_{y_1} = \frac{\partial \mathcal{L}}{\partial y_1} = -\frac{\partial V}{\partial y_1} \quad p_{y_1} = \frac{\partial \mathcal{L}}{\partial \dot{y}_1} = \frac{\partial T}{\partial \dot{y}_1}$$

What are F_{y_1} and p_{y_1} for the Atwood's machine?

1. $F_{y_1} = -mg$ and $p_{y_1} = m\dot{y}_1$
2. $F_{y_1} = -Mgy_1$ and $p_{y_1} = M\dot{y}_1$
3. $F_{y_1} = -(M - m)g$ and $p_{y_1} = (M + m)\dot{y}_1$
4. $F_{y_1} = -(M + m)g$ and $p_{y_1} = (M - m)\dot{y}_1$
5. None of these

Clicker Question 35-4

Now, we allow the pulley (mass, M_p) to rotate. The Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2}(M + m)\dot{y}_1^2 + \frac{1}{2}I\dot{\phi}^2 - (M - m)gy_1$$

Where I is the moment of inertia of the pulley. What is the moment of inertia of the pulley?

1. $I = \frac{1}{2}M_pR^2$
2. $I = \frac{1}{3}M_pR^2$
3. $I = M_pR^2$
4. $I = \frac{1}{4}M_pR^2$
5. None of these

Clicker Question 35-5

The rope moves without slipping on the pulley. A rotation of $Rd\phi$ corresponds to a displacement of dy_1 for the first mass, M . What is the **new** equation of constraint for the system?

1. $y_1 + y_2 = l - R\phi$

2. $dy_1 = Rd\phi$

3. $y_1 = R\phi$

4. $\dot{y}_1 = R\dot{\phi}$

5. More than one of these